

BACITRACIN-N OINTMENT

Ingredient(s):

Each gram contains:

Bacitracin Zinc

(eq. to Bacitracin Zinc 500 units) 12.5mg (potency)

Polymyxin B Sulfate

(eq. to Polymyxin B Sulfate 5000 units) 0.83mg (potency)

Neomycin Sulfate 3.5mg (potency)

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):

Pharmacodynamic:

Bacitracin is a bactericidal polypeptide and is active against many gram-positive bacteria including staphylococci, streptococci (particularly group A streptococci), and clostridia. It is also active against many *Actinomyces*, *Treponema pallidum*, and some gram-negative species such as *Neisseria* and *Haemophilus influenza*. Bacitracin interferes with bacterial cell wall synthesis by blocking the function of the lipid carrier molecule that transfers cell wall subunits across the cell membrane.

Neomycin is an aminoglycoside antibiotic and has bactericidal action against both gram-positive and gram-negative organisms such as staphylococci and proteus. Neomycin inhibits bacterial protein synthesis by binding to the 30S subunits of the bacterial ribosome.

Polymyxin B has a bactericidal action on most gram-negative bacilli except *Proteus* spp. It is particularly effective against *Pseudomonas aeruginosa*. Of the other gram-negative organisms, *Acinetobacter* spp., *Escherichia coli*, *Enterobacter* and *Klebsiella* spp., *Haemophilus influenza*, *Bordetella pertusis*, *Salmonella*, and *Shigella* spp. are sensitive. Polymyxin B acts primarily by binding to membrane phospholipids and disrupting the bacterial cytoplasmic membrane.

Pharmacokinetic:

Bacitracin and Polymyxin B sulfate are not absorbed to any appreciable extent from intact or denuded skin, wounds, or mucous membranes. Neomycin sulfate is not absorbed following topical application to intact skin; however, the drug is readily absorbed through denuded or abraded areas of skin or skin that has lost the keratin layer as in wounds, burns, or ulcers. Neomycin is rapidly absorbed from the peritoneum, draining sinuses, wounds, or surgical sites; use of large doses at these sites may result in substantial plasma concentrations of the drug.

Indication(s):

For the prevention or treatment of superficial skin infections caused by susceptible organisms in minor abrasions, burns, and cuts.

Dosage and Administration:

Clean the affected area. Apply a small amount on the area 1 to 3 times per day. The affected area may be covered with a sterile bandage.

Children: safety and efficacy in children younger than 2 years of age have not been established.

Contraindication(s):

Bacitracin-N ointment should not be used in individuals who have shown sensitivity to any of the ingredients.

Precaution(s) / Warning(s):

1. Due to the potential nephrotoxicity and ototoxicity of neomycin, Bacitracin-N ointment should be used with care in treating extensive burns, trophic ulcerations, or other extensive conditions where absorption is possible. Do not apply more than once daily in burn cases where more than 20% of the body is affected, especially if the patient has impaired renal function.

2. Do not use Bacitracin-N ointment in or near the eyes, nose, mouth, mucus membrane, or apply over large areas of the body.
3. Chronic application of neomycin Sulfate to inflamed skin of individuals with allergic contact dermatitis and chronic dermatoses (eg. chronic otitis external, stasis dermatitis) increases the possibility of sensitization. Low grade reddening with swelling, dry scaling, itching or a failure to heal are usually manifestation of this hypersensitivity. Discontinue use if these symptoms appear and avoid neomycin containing products thereafter.
4. Prolonged use of antibiotics may result in overgrowth of non-susceptible organism, particularly fungi. Such overgrowth may lead to a secondary infection. Discontinue the drug and take appropriate measures if superinfection occurs.

Pregnancy and Lactation:

There are no adequate and well controlled studies in pregnant women. Use only when clearly needed and when the potential benefits outweigh the unknown potential hazards to the fetus. Ototoxicity is known to occur after topical neomycin. Cranial nerve toxicity in the fetus may potentially occur with neomycin.

It is not known whether bacitracin zinc, polymixin B, or neomycin is excreted in breast milk. Exercise caution when applying on a breastfeeding woman. Neomycin has been reported to be excreted into the milk of lactating cows and ewes after a single 10mg/kg IM dose; also small amounts of other aminoglycosides are excreted into breast milk and absorbed by the nursing infant.

Side Effect(s) / Adverse Reaction(s):

Allergic contact dermatitis, hypersensitivity; ototoxicity and nephrotoxicity.

Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Keep in an airtight container. Store at temperature below 30°C. Protect from light and moisture.

Product Description and Packing(s):

A slight yellowish ointment.

Aluminium tube of 5g, 10g, and 20g.



Manufacturer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)
Lot 3, 5, 7, 12 & 14, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
43000 Kajang, Selangor, Malaysia.
Ordering Line: 1 800 88 3027
Product Info: 1 800 88 3679